

SDAF Quick Start Guide

Conducting a proportionate, evidence-based sustainability assessment

Sustainable Digital Architecture Framework — Version 1.0

1. Purpose of this guide

This Quick Start Guide provides a practical route through the Sustainable Digital Architecture Framework (SDAF) for teams undertaking their first assessment.

It explains the minimum activities and records needed to apply SDAF to a real digital architecture decision. It does not replace the complete framework or reproduce all of its requirements. Where greater methodological detail, assurance or interpretation is required, practitioners should refer to the relevant Part of SDAF v1.0.

The objective is not to produce a perfect environmental model. It is to make a better-informed decision whose scope, evidence, assumptions, uncertainty and rationale can be understood and reviewed.

2. When to use SDAF

Use SDAF when a decision could materially affect the environmental characteristics of a digital system, service or estate.

Examples include:

- selecting between on-premises, colocation, cloud or hybrid architectures;
- purchasing, leasing, retaining, refurbishing or retiring IT equipment;
- designing or modernizing applications and data platforms;
- consolidating infrastructure or migrating workloads;
- selecting technology products, suppliers or managed services;
- changing resilience, availability, capacity or performance requirements;
- extending equipment life or establishing IT asset disposition arrangements;
- comparing options with different energy, carbon, material or circularity consequences.

SDAF may be applied to a project, procurement, architecture review, transformation programme, service, product, platform or portfolio.

3. Start with the decision—not the data

Before collecting measurements, state the decision that the assessment must support.

A useful decision question identifies:

- what is being decided;
- the alternatives being considered;
- the required service or outcome;
- the relevant period;
- the affected stakeholders;
- the principal constraints.

Example

Which infrastructure option should be selected to provide the required computing service over five years while meeting performance, resilience, security, financial and sustainability requirements?

Avoid beginning with a broad question such as “Which option is greenest?” Environmental results are meaningful only when the alternatives provide sufficiently comparable functions and service outcomes.

4. Assign the minimum roles

A proportionate assessment should identify, at minimum:

- **Decision owner:** accountable for the decision and acceptance of the outcome.
- **Assessment lead:** coordinates the lifecycle and maintains the assessment record.
- **Architecture or technical lead:** confirms functional and technical comparability.
- **Evidence owners:** provide and validate relevant operational, supplier and environmental information.
- **Sustainability specialist:** advises on environmental methods where specialist interpretation is required.
- **Assurance reviewer:** conducts an independent or appropriately separated review proportionate to the decision.

One person may perform more than one role in a smaller assessment, but accountability and any limitations on independence should be recorded.

5. Determine proportionality

The assessment effort should reflect the significance and risk of the decision.

Apply greater analytical depth where the decision involves:

- high environmental or financial impact;
- long-lived or difficult-to-reverse commitments;
- material uncertainty;
- novel technologies or methods;
- weak, inconsistent or supplier-controlled evidence;
- significant stakeholder or regulatory interest;
- public environmental claims;
- major trade-offs between environmental and other requirements.

A lower-impact, reversible decision may require a lighter assessment. Proportionality may reduce the depth of analysis, but it does not remove the need to document scope, evidence, assumptions and rationale.

6. Apply the five-phase SDAF lifecycle

Phase 1 — Orient

Purpose

Establish what is being decided, why the assessment is needed and which requirements govern it.

Complete these activities

- Define the decision question.
- Identify the decision owner and assessment participants.
- Describe the required function, service and outcomes.
- Identify applicable architectural, sustainability, financial, risk, legal and policy requirements.
- List the alternatives to be evaluated.
- Determine the required assessment depth and assurance level.
- Record exclusions, constraints and potential conflicts between objectives.

Minimum output

An approved assessment brief containing the decision question, scope, alternatives, stakeholders, applicable requirements and proportionality determination.

Check before proceeding

Can every participant explain what decision the assessment will support?

If not, refine the decision question before gathering data.

Phase 2 — Baseline

Purpose

Establish a consistent reference against which alternatives can be evaluated.

Complete these activities

- Define the system boundary.
- Define the Functional Unit: the quantified function or service being provided.
- Define the Reference Flow: the quantity of technology, infrastructure or activity required to deliver that function.
- Describe the current state or reference option.
- Establish the assessment period and relevant lifecycle stages.
- Identify existing assets, dependencies, utilization, capacity and operating conditions.
- Create an evidence register.
- Record data sources, ownership, dates, assumptions, gaps and limitations.

Minimum output

A documented baseline, system boundary, Functional Unit, Reference Flow and initial evidence register.

Check before proceeding

Are the alternatives being compared on the basis of the same function and an equivalent level of service?

If not, normalize the comparison or document why direct comparison is not possible.

Phase 3 — Evaluate*Purpose*

Generate and interpret sufficient evidence to understand the material consequences of each alternative.

Complete these activities

- Select relevant environmental indicators.
- Evaluate applicable lifecycle stages rather than operational energy alone.
- Consider embodied and use-phase impacts.
- Examine energy, emissions, materials, water and waste where material.
- Evaluate equipment life, repairability, reuse, refurbishment, recovery and other circularity considerations.
- Test important assumptions and uncertain inputs.
- Assess evidence quality, completeness, representativeness and traceability.
- Identify trade-offs with cost, risk, security, resilience, performance and service quality.
- Avoid combining unlike indicators into a single score unless the aggregation method and value judgements are explicit.

Minimum output

Comparable results for the alternatives, supported by documented sources, assumptions, uncertainty, limitations and evidence-quality findings.

Check before proceeding

Would a reasonable reviewer be able to understand how the results were produced and which uncertainties could affect them?

If not, strengthen the evidence or narrow the conclusions.

Phase 4 — Architect*Purpose*

Use the evidence to select and document a defensible course of action.

Complete these activities

- Confirm that the alternatives remain functionally and technically viable.
- Test whether the evidence is sufficient for the significance of the decision.
- Identify the preferred option and material trade-offs.
- Determine whether uncertainty could change the preferred option.
- Apply the appropriate SDAF decision class.
- Record conditions, mitigations and implementation requirements.
- Define circumstances that would cause the decision to be reconsidered.
- Obtain approval from the accountable decision owner.

Minimum output

A decision record explaining:

- the option selected;
- the alternatives considered;
- the evidence relied upon;
- the material trade-offs;
- the treatment of uncertainty;
- the decision class;
- the approval and rationale;
- any conditions or review triggers.

Check before proceeding

Does the decision record explain why the selected option is justified—not merely which option produced the lowest reported figure?

Phase 5 — Assure*Purpose*

Confirm that the assessment was conducted appropriately and that its conclusions are supported by the available evidence.

Complete these activities

- Review conformity with applicable SDAF requirements.
- Check the scope, boundary, Functional Unit and Reference Flow.
- Verify important sources, calculations, transformations and assumptions.
- Review the evidence-quality and uncertainty determinations.
- Check for unsupported claims, omissions and double counting.
- Confirm that the decision rationale is consistent with the evidence.
- Record findings, limitations and corrective actions.
- Establish monitoring measures, ownership and review dates.

Minimum output

An assurance record containing the review scope, findings, conclusion, corrective actions, limitations and validity period.

Check before closure

Are the assessment record, decision record and assurance findings complete, traceable and accessible to authorized reviewers?

7. Assemble the minimum SDAF assessment record

A first SDAF assessment should create one controlled record containing:

1. assessment title and identifier;
2. decision question;
3. decision owner and assessment roles;
4. scope and system boundary;
5. Functional Unit and Reference Flow;
6. current-state baseline;
7. alternatives evaluated;
8. applicable requirements and constraints;
9. evidence register;
10. methods, calculations and transformations;
11. assumptions, exclusions and data gaps;
12. evidence-quality and uncertainty determinations;
13. lifecycle and circularity findings;
14. comparison of alternatives and material trade-offs;
15. selected decision class;
16. decision and rationale;
17. approvals and conditions;
18. assurance findings;
19. monitoring measures and review triggers;
20. version history.

Existing architecture, procurement, risk or governance records may be used. SDAF does not require duplicate documentation where the required information is already controlled and traceable.

8. Use evidence with appropriate confidence

Not all data carries the same evidential weight.

For each material input, ask:

- Is its source identifiable?
- Is it current enough for the decision?
- Does it represent the relevant technology, geography and operating conditions?
- Is the method documented?
- Can the result be reproduced?
- Has uncertainty been stated?
- Is the information independently verified or dependent on an untested claim?
- Could a plausible change in the input alter the decision?

Do not convert weak evidence into apparent certainty through additional calculation. Where evidence is limited, state the limitation and reduce the strength, duration or scope of the conclusion.

9. Avoid these common errors

- Starting with available data rather than the decision question.
- Comparing products or services that do not provide equivalent functions.
- Treating operational energy as the whole lifecycle impact.
- Using supplier claims without examining their scope and methodology.
- Mixing organizational emissions inventories with product or decision-level assessments.
- Counting circularity benefits more than once.
- Hiding data gaps inside aggregated scores.
- Selecting an option solely because one environmental indicator is lower.
- Treating governance approval as permission to override SDAF methodology.
- Treating an assurance finding as proof that an option is inherently sustainable.
- Reusing a previous assessment without checking whether its context remains valid.

10. Know when specialist support is required

Seek appropriate technical or sustainability expertise when the assessment requires:

- a formal lifecycle assessment or product carbon footprint;
- complex allocation or avoided-burden modelling;
- interpretation of Environmental Product Declarations;
- Scope 1, Scope 2 or Scope 3 accounting conclusions;
- materiality determinations for public reporting;
- regulatory or contractual conformance;
- independent verification;
- environmental claims intended for external publication.

SDAF structures the decision and its evidence. It does not remove the need for competent application of specialist standards and methods.

11. Close the assessment with a clear decision statement

A complete SDAF decision statement should answer:

What was decided, on whose authority, using which scope and evidence, with what level of confidence, under which conditions, and until when does the conclusion remain valid?

If those elements cannot be stated clearly, the assessment is not ready to close.

Further information

The complete normative requirements, explanatory guidance and supporting examples are contained in:

Sustainable Digital Architecture Framework (SDAF): An Integrated Framework for Evidence-Based, Defensible and Sustainable Architectural Decisions — Version 1.0

Framework: <https://doi.org/10.5281/zenodo.22899663>

Templates and implementation resources: <https://github.com/Raymond-du-Toit/SDAF>

Feedback and community review: <https://github.com/Raymond-du-Toit/SDAF/blob/main/CONTRIBUTING.md>

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